

3<sup>rd</sup> Sem.-B.Tech.

END-SEMESTER EXAMINATION, NOV.-DEC.,2023

Course Code- EAMTC04

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

| Q.No. | Attempt any 2 parts of the questions below                                                                                                                                         | Marks | CO  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| 1(a)  | Find KerT & ImT for the linear mapping $T: R^2 \rightarrow R^3$ defined by $T(x, y) = (x + y, x - y, y)$ .                                                                         | 05    | CO1 |
| 1(b)  | Show that $T: R^4 \rightarrow R^4$ $T(x, y, z, t) = (2x, 3y, 0, 0)$ is a linear transformation. Find its rank and nullity.                                                         | 05    | CO1 |
| 1(c)  | Find a basis and its dimension for the subspace of $R^3$ spanned by the set $S = \{(3, 2, 1), (1, 2, 0), (-1, 2, -1)\}$                                                            | 05    | CO1 |
| 2(a)  | Use the Gram-Schmidt process to convert the basis $\{(1, 0, 1), (1, 0, -1), (0, 3, 4)\}$ of the Euclidean space $R^3$ with standard inner product into an orthonormal basis.       | 05    | CO2 |
| 2(b)  | Find the eigen values, eigen vectors and eigen spaces of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                                                                        | 05    | CO2 |
| 2(c)  | Reduce the quadratic form $5x^2 + y^2 + 10z^2 - 4yz - 10zx$ to the normal form and show that it is positive definite.                                                              | 05    | CO2 |
| 3(a)  | Find the polar decomposition of $A = \begin{pmatrix} 11 & -5 \\ -2 & 10 \end{pmatrix}$                                                                                             | 05    | CO3 |
| 3(b)  | Find the singular value decomposition of the matrix $A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$                                                                    | 05    | CO3 |
| 3(c)  | Find Pseudo inverse of the matrix $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix}$                                                                                     | 05    | CO3 |
| 4(a)  | Verify Green's Lemma in the xy plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the closed curve of the region bounded by $y = \sqrt{x}, y = x^2$ .                     | 05    | CO4 |
| 4(b)  | Find $\text{div} \vec{F}$ and $\text{curl} \vec{F}$ where $\vec{F} = \text{grad} (x^3 + y^3 + z^3 - 3xyz)$                                                                         | 05    | CO4 |
| 4(c)  | If $\vec{F} = x\hat{i} + (z^2 - zx)\hat{j} - xy\hat{k}$ . Evaluate $\iint_S \vec{F} \cdot \vec{ds}$ where S is the surface of the triangle with vertices (2,0,0), (0,2,0), (0,0,4) | 05    | CO4 |
| 5(a)  | Examine if the set $S = \{(x, y) \in R^2: x^2 + y^2 \leq 4\}$ is a convex set.                                                                                                     | 05    | CO5 |
| 5(b)  | Minimize $x$<br>Subject to $x^2 + y^2 = 1$                                                                                                                                         | 05    | CO5 |
| 5(c)  | Solve: Min $f(x) = -2x_1 - x_2$<br>subject to $x_1 - x_2 \leq 0$<br>$x_1^2 + x_2^2 \leq 4$<br>$x_1, x_2 \geq 0$                                                                    | 05    | CO5 |